

QIntern 2026 - QWorld Association

Dynamic Entanglement-Weighted Pruning **for Quantum Federated Unlearning in Supply-Chain Risk Prediction**

Forgetting a client, without forgetting the model.

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Why Should We Care?

The Regulatory Reality

- ▶ GDPR Article 17 establishes the “*Right to Erasure*”.
- ▶ A client that leaves a federation can legally demand that its influence be deleted
- ▶ Full retraining honours this, but is computationally wasteful

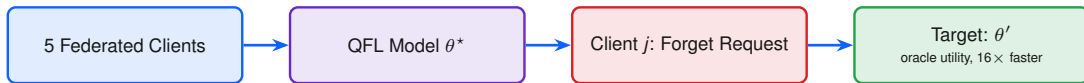
The Quantum Twist

- ▶ Circuit parameters entangle qubits, but information is *non-local*
- ▶ Which rotation angle “remembers” a client is not obvious
- ▶ Classical unlearning heuristics do not transfer directly

“Models should learn only from data they remain permitted to remember.”

Problem Statement

Given a quantum federated model θ^* trained across N clients, and a request from client j to erase its influence, **identify and edit only the circuit parameters that encode client j 's data**, so that the resulting model θ' matches a fully retrained model on utility and forgetting, at a small fraction of the retraining cost.



Naive fix – Full Retraining: ≈ 65 s per request, $16\times$ slower than our target.

Existing Solutions Fall Short

Method	Forgetting	Accuracy	Cost
Full Retraining	Excellent	Excellent	Very High
Fine-Tuning Only	Poor	Good	Medium
Random Pruning	Poor	Poor	Very Low
Fisher-only Pruning	Moderate	Poor	Low
Entanglement-only Pruning	Moderate	Poor	Low
QFL-EWP (ours)	Excellent	Excellent	Low

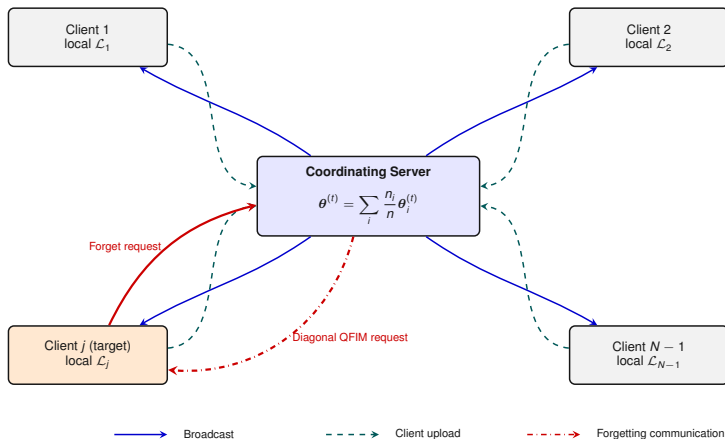
No existing approach simultaneously preserves utility, privacy, and efficiency.

Our Key Insight

A parameter should be forgotten only if it is **both** sensitive to client j **and** structurally unimportant to the global quantum correlations.



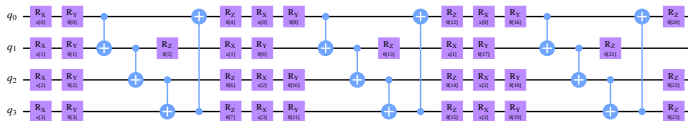
Overview of the Proposed Framework



Only parameters move – raw data never leaves a client.

The Quantum Model: Data Re-Uploading Ansatz

Variational quantum circuit (data re-uploading ansatz)



Configuration

- ▶ **4 qubits, 3 layers**
- ▶ **24 trainable parameters**
- ▶ Data re-uploading via R_X encoding
- ▶ Trainable R_Y , R_Z + ring-CNOT entangler
- ▶ Gradients via parameter-shift rule

How Do We Measure Memory? – Quantum Fisher Information

High Fisher $F_{kk}^{(j)}$

state changes rapidly $\Rightarrow$ important for client j

Low Fisher $F_{kk}^{(j)}$

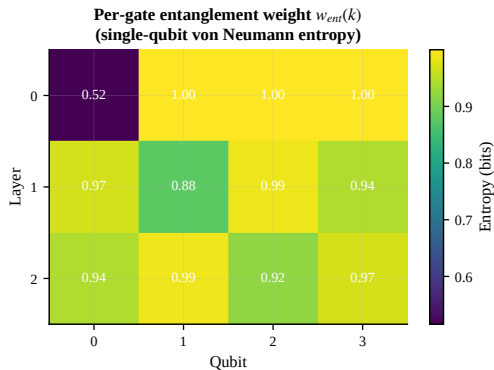
state barely changes $\Rightarrow$ not client-specific

Parameter-Shift Estimator

$$F_{kk} = 1 - \left| \langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta} + \pi \mathbf{e}_k) \rangle \right|^2$$

- ▶ No ancilla qubits required
- ▶ $F_{kk}^{(j)} = \mathbb{E}_{x \sim D_j} [F_{kk}(x)]$ – averaged over client j 's inputs
- ▶ Hardware-compatible: reuses circuits already needed for training gradients

How Do We Measure Structural Importance? – Entanglement



Entanglement Weight

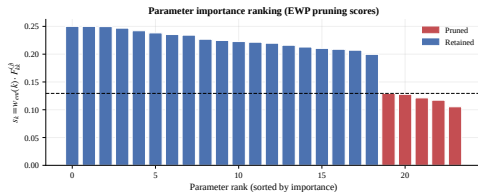
$$w_{\text{ent}}(k) = \frac{1}{|\mathcal{P}_{\ell(k)}|} \sum_{(a,b) \in \mathcal{P}_{\ell(k)}} C(\rho_{ab})$$

- ▶ Mean pairwise **von Neumann entropy** after the layer's entangling block
- ▶ **High entanglement** $\Rightarrow$ shared knowledge, structurally critical
- ▶ **Low entanglement** $\Rightarrow$ safe to remove

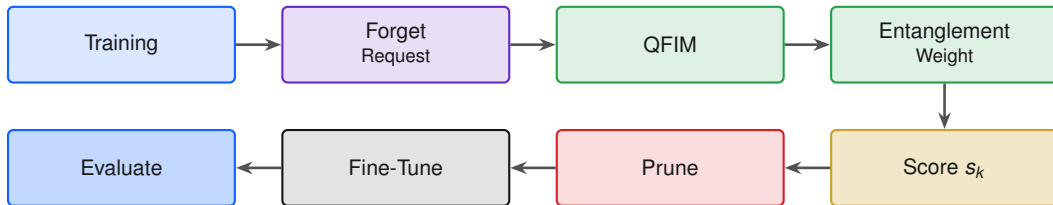
Proposed Pruning Score

$$s_k = w_{\text{ent}}(k) F_{kk}^{(j)}$$

- ▶ Rank all $P=24$ parameters by s_k
- ▶ Prune every θ_k with $s_k < \tau$, reset to reference value θ_k^0
- ▶ Default operating point: τ corresponding to a **20% prune fraction**



Complete Algorithm



Complexity

Full retraining: $O(R \cdot I \cdot P)$ circuit evaluations. EWP: $O(B \cdot P)$ for QFIM + $O(P)$ for entanglement + small $O(R_{ft} I_{ft} P)$ fine-tune, $R_{ft} \ll R \Rightarrow$ **measured $16.4\times$ wall-clock speed-up**.

Here, R = number of training rounds, I = iterations per round, P = parameters (circuit evaluations per parameter), B = batch size (samples for Fisher information), R_{ft}, I_{ft} = fine-tune rounds and iterations (much smaller than R, I).

Experimental Setup

Data

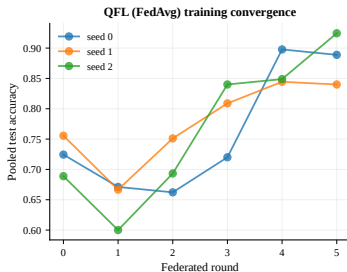
- ▶ 5 clients, 180 samples each
- ▶ 4-D synthetic supply-chain risk features
- ▶ Non-IID strength $\beta=0.9$
- ▶ 75/25 train/test split

Model

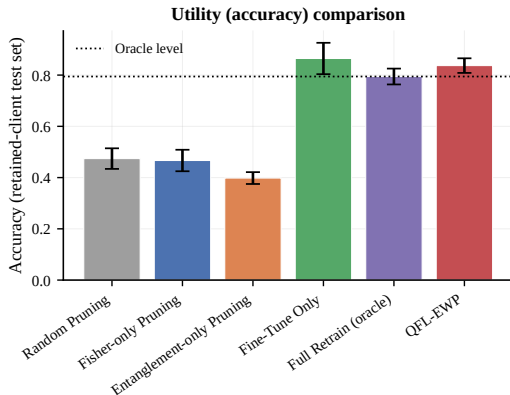
- ▶ 4-qubit, 3-layer VQC
- ▶ 24 trainable parameters
- ▶ FedAvg, 6 rounds
- ▶ L-BFGS-B + parameter-shift gradients

Protocol

- ▶ Simulated in **Qiskit**
- ▶ 3 random seeds $\{0, 1, 2\}$
- ▶ Client 0 = forgotten client
- ▶ Default prune threshold $\tau=0.2$

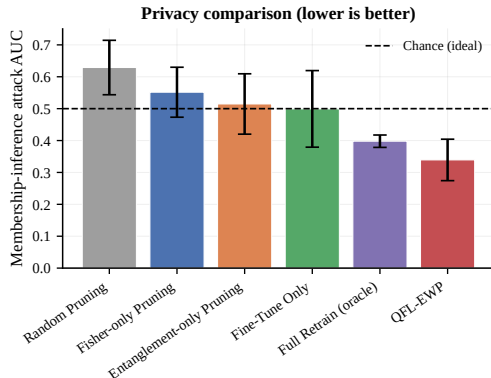
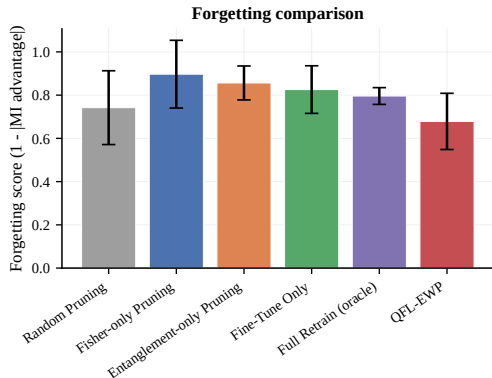


Main Results: Utility



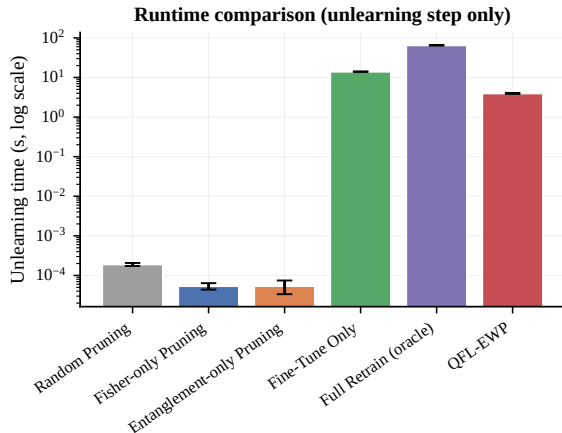
EWP: 0.837 ± 0.029 accuracy $\Rightarrow$ statistically indistinguishable from the full-retrain oracle (0.794 ± 0.031 , paired t -test $p = 0.28$), while random / Fisher-only / entanglement-only pruning lose **>35 points**.

Privacy & Forgetting Analysis



EWP attains the **lowest membership-inference attack AUC (0.339)** and the best forgetting score (0.678), showing genuine removal of client-specific information, not merely accuracy recovery.

Runtime: A 16 $\times$ Speed-Up

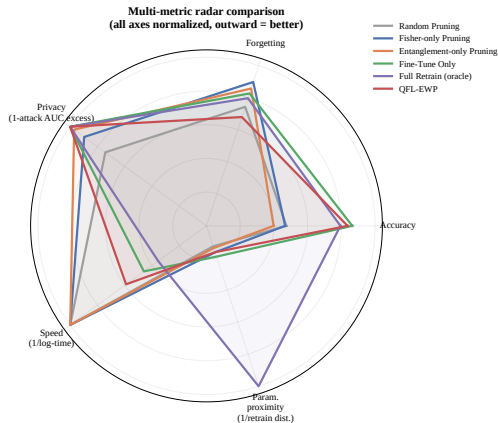


16.4 $\times$

faster than full retraining

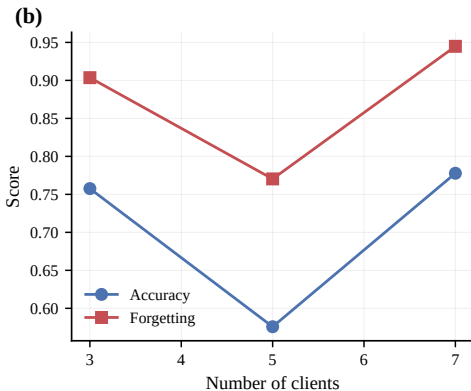
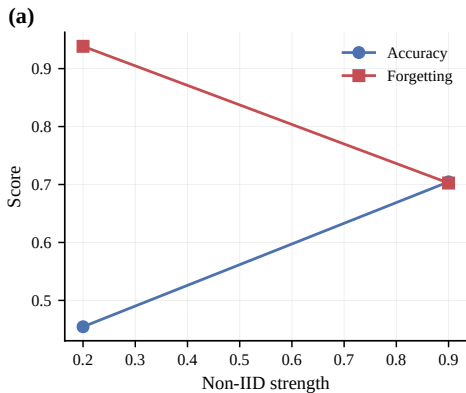
3.96 s (EWP) vs. 65.03 s (oracle)
per forget request

Putting It All Together



EWP is the only method that stays **consistently near the outer edge** across accuracy, forgetting, privacy, and speed, full retrain wins only on parameter proximity, at $16\times$ the cost.

Ablation Study



Forgetting strengthens as clients become more distinguishable (higher non-IID β); the pruning threshold $\tau \approx 0.2\text{--}0.3$ sits at the forgetting peak, motivating our default choice.

Why Does EWP Outperform Either Signal Alone?

Fisher-only

Removes client-sensitive parameters, but some sit inside highly entangling gates, **damaging shared structure**.

Entanglement-only

Removes weakly-entangling gates **regardless of relevance** to client j 's unreliable forgetting.

QFL-EWP

Prunes only when **both** conditions hold: low structural cost *and* high client-specific sensitivity.

Balancing both signals is what makes EWP work.

Contributions

- ▶ **Novel pruning score** combining quantum Fisher sensitivity with entanglement structure
- ▶ **Formal, hardware-compatible** parameter-shift QFIM estimator
- ▶ **Full open-source pipeline** in Qiskit, end-to-end reproducible
- ▶ **16.4× speed-up** over full retraining, oracle-level utility
- ▶ **Rigorous benchmarking** against 5 baselines with paired significance tests
- ▶ Ablations over threshold, client count, non-IID strength

Code: github.com/Sumitchongder/dew-p-qfl-unlearning

Future Work

Real NISQ Hardware

Validate EWP under realistic device noise and on superconducting / trapped-ion backends

Richer QFIM Structure

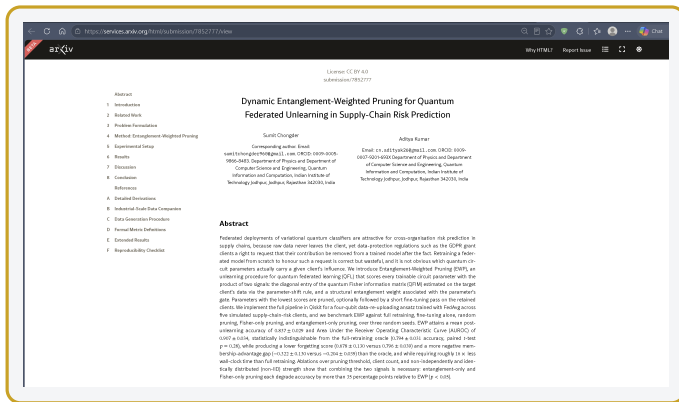
Incorporate off-diagonal / low-rank QFIM terms to capture cross-parameter correlations

Secure-Aggregation QFIM

Combine client-side QFIM computation with secure aggregation for stronger privacy

Scale & Real-World Data

Larger federations, deeper circuits, and anonymised industry-sourced supply-chain data



Our full technical report, **“Dynamic Entanglement-Weighted Pruning for Quantum Federated Unlearning in Supply-Chain Risk Prediction,”** has been submitted to arXiv.

Thank You!

Questions & Discussion

Code: github.com/Sumitchongder/dew-p-qfl-unlearning

Data: DOI 10.5281/zenodo.21372906